

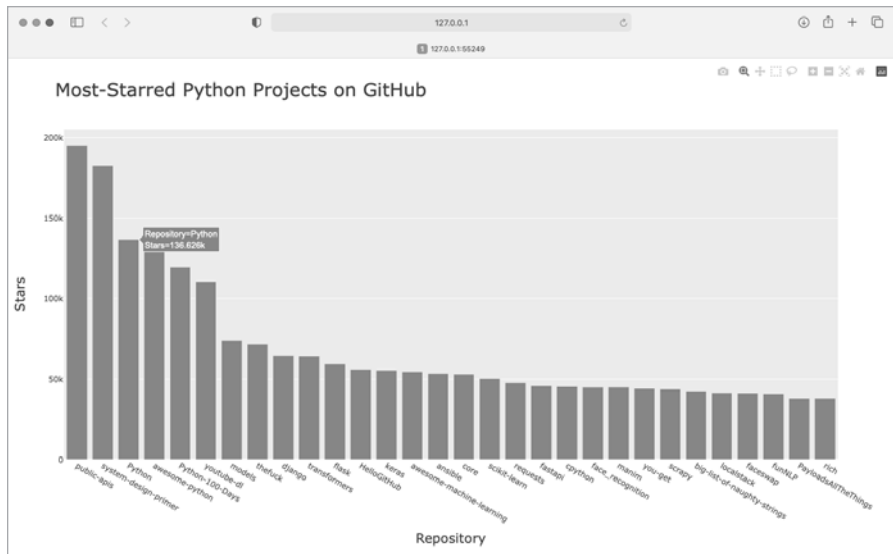


Figure 17-2: A title has been added to the main chart, and to each axis as well.

Adding Custom Tooltips

In Plotly, you can hover the cursor over an individual bar to show the information the bar represents. This is commonly called a *tooltip*, and in this case, it currently shows the number of stars a project has. Let's create a custom tooltip to show each project's description as well as the project's owner.

We need to pull some additional data to generate the tooltips:

```
python_repos
_visual.py

--snip--
# Process repository information.
repo_dicts = response_dict['items']
❶ repo_names, stars, hover_texts = [], [], []
for repo_dict in repo_dicts:
    repo_names.append(repo_dict['name'])
    stars.append(repo_dict['stargazers_count'])

    # Build hover texts.
    ❷ owner = repo_dict['owner']['login']
    description = repo_dict['description']
    ❸ hover_text = f"{owner}<br />{description}"
    hover_texts.append(hover_text)

# Make visualization.
title = "Most-Starred Python Projects on GitHub"
labels = {'x': 'Repository', 'y': 'Stars'}
❹ fig = px.bar(x=repo_names, y=stars, title=title, labels=labels,
               hover_name=hover_texts)
```